

# $T\bar{T}$ deformation in large N limit

Junichi Haruna  
Kyoto University

work in progress with T.Ishii, H.Kawai, K.Sakai, K.Yoshida

## 0. Summary

- We studied infinitesimal  $T\bar{T}$  deformation of  $O(N)$  massive vector model in large N limit:

$$\frac{\mathcal{L}}{N} = \frac{1}{2} (\partial_\mu \vec{\phi})^2 - \frac{m_0^2}{2} \vec{\phi}^2 + \alpha_0 \det(T_{\mu\nu})$$

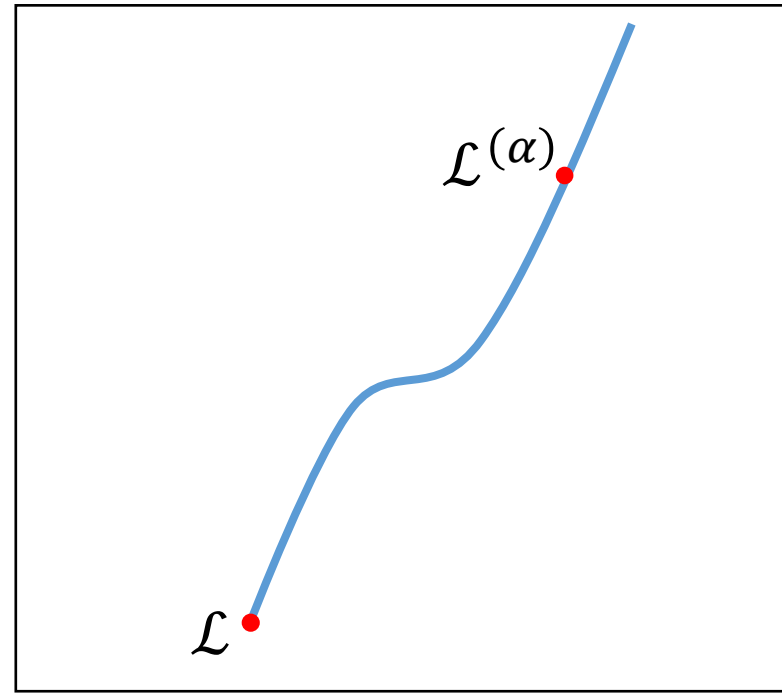
- We calculated effective action at  $O(1/N)$  (sub-leading) level.
- The effective action breaks the positivity of state space.
- Local symmetries such as diffeomorphism invariance will recover the positivity.

## 1. Introduction

- $T\bar{T}$  deformation is an irrelevant deformation of 2 dimensional Lagrangian with the  $T\bar{T}$  flow equation<sup>[1]</sup>:

$$\frac{\partial \mathcal{L}^{(\alpha)}}{\partial \alpha} = \det(T_{\mu\nu}^{(\alpha)})$$

$$\left( \begin{array}{l} \alpha \dots \text{deformation parameter} \\ \mathcal{L}^{(\alpha)} \dots \text{deformed Lagrangian} \\ T_{\mu\nu}^{(\alpha)} \dots \text{Energy-Momentum tensor of } \mathcal{L}^{(\alpha)} \end{array} \right)$$



- $T\bar{T}$  deformation is equivalent to a classical metric perturbation with field-dependent coordinate transformation.<sup>[2]</sup>
- If the  $T\bar{T}$  flow can be understood at quantum level, a clue to quantum gravity may be captured.
- We study infinitesimal  $T\bar{T}$  deformation of  **$O(N)$  massive vector model in large N limit at quantum level.**

## 2. Effective action

- The Lagrangian can be rewritten with auxiliary fields:

$$\begin{aligned} \frac{\mathcal{L}}{N} &= \frac{1}{2} (\partial_\mu \vec{\phi})^2 - \frac{m_0^2}{2} \vec{\phi}^2 - \frac{1}{2} C T_\mu^\mu - \frac{1}{2} \tilde{C}^{\mu\nu} \tilde{T}_{\mu\nu} - \frac{1}{4\alpha_0} C^2 + \frac{1}{8\alpha_0} (\tilde{C}^{\mu\nu})^2 \\ &= \frac{1}{2} \vec{\phi} \cdot (-\partial^2 - (1-C)m_0^2 + \partial_\mu \tilde{C}^{\mu\nu} \partial_\nu) \vec{\phi} - \frac{1}{4\alpha_0} C^2 + \frac{1}{8\alpha_0} (\tilde{C}^{\mu\nu})^2 \end{aligned}$$

$$\left( \begin{array}{l} \tilde{C}^{\mu\nu} \dots \text{symmetric traceless tensor field} \\ C \dots \text{scalar field} \end{array} \right)$$

- The effective action:

$$\frac{S_{\text{eff}}}{N} = \frac{i}{2} \text{tr} \log (-\partial^2 - (1-C) + \partial_\mu \tilde{C}^{\mu\nu} \partial_\nu) + \int_x -\frac{1}{4\alpha_0} C^2 + \frac{1}{8\alpha_0} (\tilde{C}^{\mu\nu})^2$$

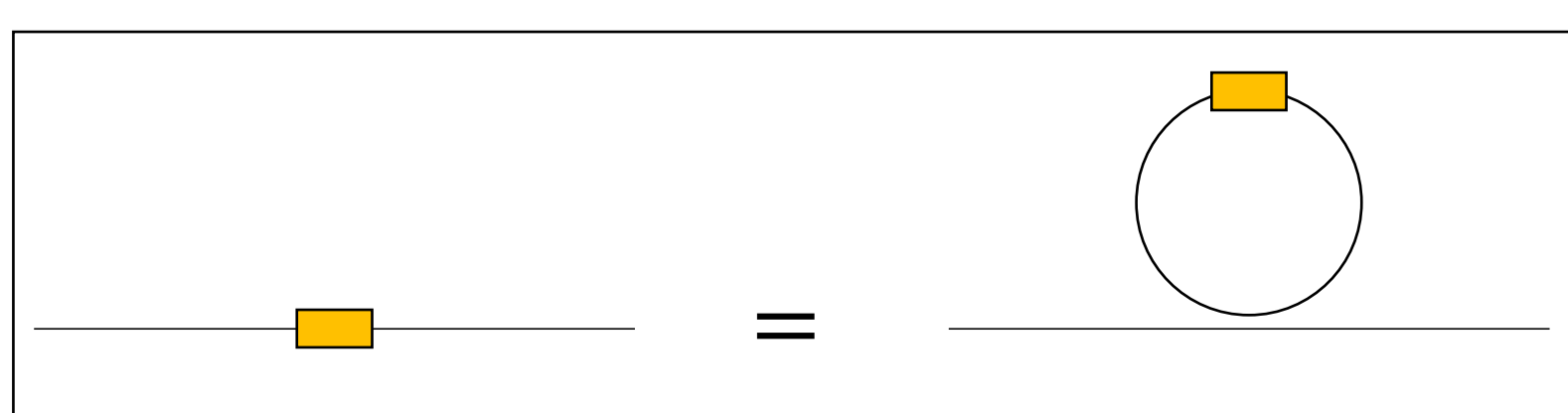
- We consider leading and sub-leading ( $O(1/N)$ ) contributions to the effective action.

### 2-1. Leading contribution

- In  $N \rightarrow \infty$  limit,  $C$  and  $\tilde{C}^{\mu\nu}$  integral is equivalent to substituting the value of stationary point of  $S_{\text{eff}}$ .

- Stationary condition:

$$\left( \begin{array}{l} \frac{\delta S_{\text{eff}}}{\delta \tilde{C}^{\mu\nu}} = 0 \\ \frac{\delta S_{\text{eff}}}{\delta C} = 0 \end{array} \right) \Leftrightarrow \left( \begin{array}{l} \langle \tilde{C}^{\mu\nu} \rangle = 0 \\ \frac{\langle C \rangle}{\alpha_0 m_0^2} = \int_k \frac{i}{k^2 - (1 - \langle C \rangle) m_0^2} \end{array} \right)$$



- The **logarithmic divergence** of the loop integral can be **renormalized** into  $\alpha_0 m_0^2$ .

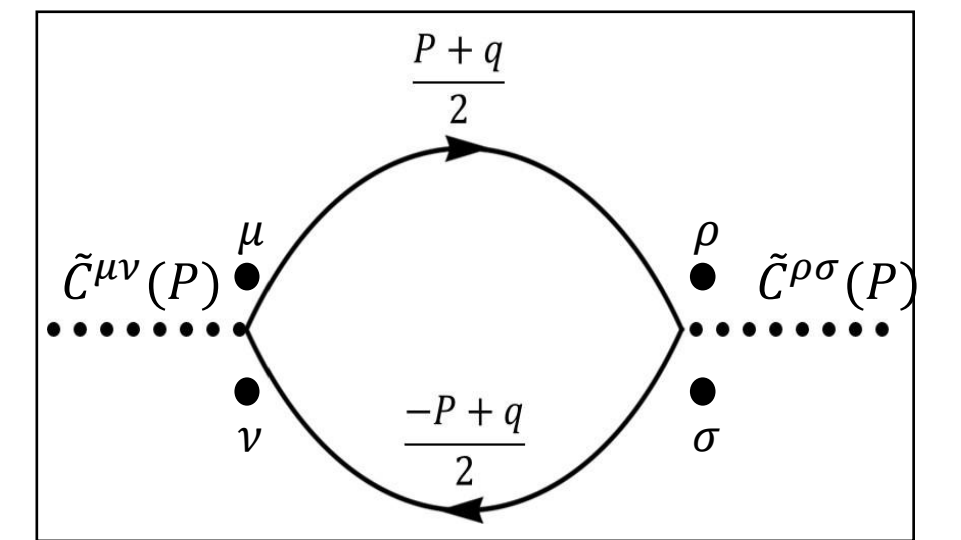
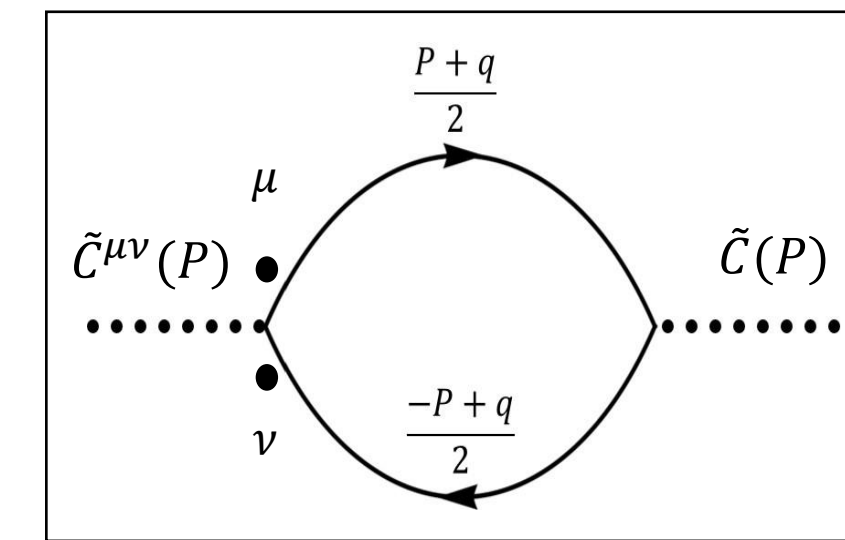
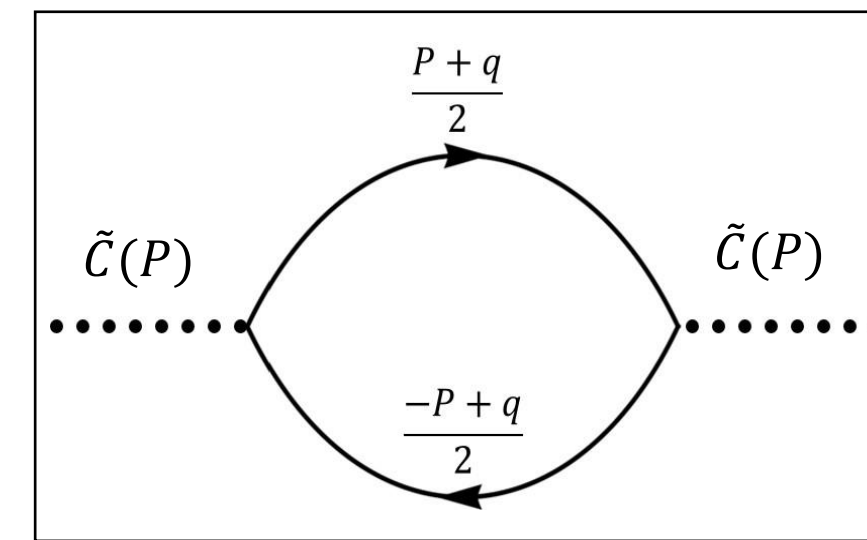
[Note:  $\langle C \rangle \neq 0$  does not mean spontaneous breaking of  $O(N)$  symmetry because  $\langle C \rangle \sim \langle \vec{\phi}^2 \rangle$ .]

### 2-2. Sub-leading contribution

- To consider  $O(1/N)$  correction to the effective action,

We set  $C = \langle C \rangle + \frac{m^2}{m_0^2 \sqrt{N}} \tilde{C}$ ,  $\tilde{C}^{\mu\nu} \rightarrow \frac{\tilde{C}^{\mu\nu}}{\sqrt{N}}$  and expand  $S_{\text{eff}}$  up to 2<sup>nd</sup> order of  $\tilde{C}$ ,  $\tilde{C}^{\mu\nu}$ :

$$\begin{aligned} S_{\text{eff}} &= \frac{1}{2} \int_P \left( a P^2 - \frac{1}{4\alpha_0} + (\text{quadratic div.}) \right) (\tilde{C}^{\mu\nu})^2 + \\ &\quad \left( \frac{m^4}{4\pi P \sqrt{4m^2 - P^2}} \tan^{-1} \left( \frac{P}{\sqrt{4m^2 - P^2}} \right) - \frac{P^2}{96\pi} - \frac{m^2}{16\pi} \right) \left( \tilde{C} - \frac{P_\mu P_\nu}{P^2} \tilde{C}^{\mu\nu} \right)^2 \\ &\quad \left( a \dots \text{scheme-dependent constant} \right. \\ &\quad \left. \text{e.g. } 0 \text{ (Pauli-Villars, Dimensional)}, -\frac{1}{12\pi} \text{ (Cutoff)} \right) \end{aligned}$$



#### Renormalizability

- The quadratic divergence can be renormalized into  $\alpha_0$ .  
 $\Rightarrow$  **All divergence can be removed** up to  $O(1/N)$ .

- Finite renormalization terms in  $S_{\text{eff}}$  correspond to **non-local counterterms** in the original Lagrangian  $\mathcal{L}$ :

$$S_{\text{finite ren.}} = \int_x \frac{1}{2} \tilde{C}^{\mu\nu} \delta_Z \partial^2 \tilde{C}_{\mu\nu} \leftrightarrow \frac{\mathcal{L}_{\text{c.t.}}}{N} = \frac{\alpha_0^2}{2} \tilde{T}^{\mu\nu} \frac{\delta_Z \partial^2}{1 + \alpha_0 \delta_Z \partial^2} \tilde{T}_{\mu\nu}$$

#### Positivity

- In UV ( $P^2 \rightarrow \infty$ ) region, the kinetic term of  $\tilde{C}$  has wrong sign.
- If  $a \neq 0$ , because  $(\tilde{C}^{\mu\nu})^2 = (\tilde{C}^{00})^2 - (\tilde{C}^{01})^2$ , the kinetic term of either  $\tilde{C}^{00}$  or  $\tilde{C}^{01}$  has wrong sign.  
 $\Rightarrow$  **Positivity of state space is violated.**

## 3. Discussion

- The effective action is essentially same as Liouville action

$$S_L = -\frac{N-25}{24\pi} \int_x \sqrt{-g} \left( \frac{1}{2} g^{\mu\nu} \partial_\mu \sigma \partial_\nu \sigma + R\sigma \right)$$

$$\text{regarding } g^{\mu\nu} \sim \left( 1 + \frac{\tilde{C}}{2\sqrt{N}} \right) \eta^{\mu\nu} + \frac{\tilde{C}^{\mu\nu}}{2\sqrt{N}}$$

without FP ghost contribution ( $N-25 \rightarrow N+1$ )

- Liouville action with  $N \leq 25$  does not violate the positivity because of **diffeomorphism invariance** and its FP ghost.
- The above results are valid for general deformation by any function of  $T_{\mu\nu}$  with a slight modification.
- If deformation of 2D theory by any function of  $T_{\mu\nu}$  is well-defined at quantum level, **the deformed theory should have local symmetries such as diffeomorphism.**

[This conclusion is consistent with the fact that  $a=0$  with renormalization schemes which hold diffeomorphism invariance (Pauli-Villars and Dimensional Regularization).]

## References

- [1] F.A.Smirnov and A.B.Zamolodchikov, arXiv:1608.05499
- [2] S.Dubovsky, V.Gorbenko and M.Mirbabayi arXiv:1706.06604